Chapter 7

CORRELATION ANALYSIS

7.1 INTRODUCTION

The statistical methods like measures of central tendency, dispersion, skewness etc. discussed so far are used to analyse or compare the data involving one variable only. But often an analysis of data concerning two or more variables is needed to describe the statistical relationship or association between them e.g. relation of income and expenditure, sales and consumption, height and weight etc. For this, the correlation analysis is one of the most common and useful statistical method to evaluate the strength and direction of relationship between two variable in a single figure.

Sir Francis Galton was the first statistician who described the concept and developed the technique to show correlation graphically in 1896, but Karl Pearson developed the proper technique to find correlation based on mathematical calculations.

7.2 MEANING AND DEFINITION

A statistical technique used to analyse the strength and direction of the relationship between two variables is *correlation analysis* and the measure of correlation in a single figure is called *coefficient of correlation.* It shows the extent of linear relationship between them i.e. higher the correlation coefficient, stronger the relationship.

(i) The strength of relationship is determined by the closeness of the points to a straight line when a pair of values of two variables are plotted on a graph.

(ii) The direction is determined by whether a change in the value of one variable is accompanied by a change in the value of the other variable, i.e.

(a) an increase (or decrease) in one variable is accompanied by an increase (or decrease) in other.

(b) an increase (or decrease) in one variable is accompanied by decrease (or increase) in other.

Some definitions of correlation are given below:

(1) "Correlation is an analysis of the co-variation between two or more variables." **–A.M. Tuttle**

(2) "The effect of correlation is to reduce the range of uncertainty of one's prediction." **–Tippett**

(3) "If two or more quantities vary in sympathy, so that movement in one tend to be accompanied by corresponding movements in the other, then they are said to be correlated." **–Connor**

(4) "Correlation analysis deals with the association between two or more variables." **–Simpson and Kafka**

(5) "Correlation analysis attempts to determine the degree of relationship between variables." **–Ya Lun Chou**

(6) "When the relationship is of a quantitative nature, the appropriate statistical tool for discovering and measuring the relationship and expressing it in a brief formula is known as correlation." **–Croxton and Cowden**

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7.6 TYPES OF CORRELATION

Correlation is classified in several different ways. The three most important types are:

(1) Positive, Negative and Neutral
(2) Simple, Multiple and Partial
(3) Linear and Non-Linear

1. Positive and Negative Correlation

On the basis of direction in which the variables move, the correlation can be classified into two types.

(a) Positive or Direct Correlation : When the values of two variables move in the same direction i.e. if one increases then the other also increases and vice-versa, then it is called the *positive correlation* e.g. price and supply, income and expenditure, amount of rainfall and yield of crops etc.

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(b) Negative or Indirect Correlation : When the values of two variables move in the opposite direction *i.e.* if one increases then the other decreases and vice-versa, then it is called *negative correlation* e.g. price and demand, investment and rate of interest etc.

X:	4	6	8	10	12	X:	4	6	8	10	12
Y:	5	10	15	20	25	Y:	25	20	15	10	5
Positive Correlation						Negative Correlation					

Neutral Correlation or No correlation : When there is no relationship in the change of the variables then it is said to be neutral correlation. It can also be called as zero correlation.

2. Simple, Multiple and Partial Correlation

On the basis of number of variables studied, the correlation may be classified into three types:

(a) Simple Correlation : If we study correlation between the two variables only, then such a correlation is referred to as simple correlation. e.g. height and weight, income and consumption etc.

(b) Multiple correlation : If we study correlation between more than two variables simultaneously then it is called *multiple correlation* e.g. Study of relation between marks of English, Economics, Statistics simultaneously, or the study of relationship between rainfall, yield of rice, amount of fertilizers etc.

(c) Partial Correlation : When there are more than two variables, but we study the relationship between only two of them, assuming other variables as constant, is called partial correlation. e.g. the study of relationship between rainfall and crops, without taking into consideration effects of other factors like fertilizers, seeds, pesticides etc.

3. Linear and Non-Linear Correlation

On the basis of change in proportion of variables under study, the correlation may be classified as:

(a) Linear Correlation : When the change in the values of two variables is either proportional or fixed, then they are said to be *linearly correlated* i.e. corresponding to a unit change in one variable, there is a constant change in the other variable over the whole set of values. If such variables are plotted on a graph paper, we will find that the plotted points will cluster around a straight line, and this line is known as best fitted line, and expressed as $Y = a + bX$, where a, b are constants.

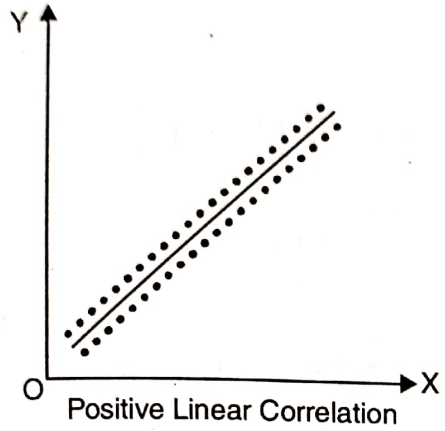
(b) Non-Linear Correlation : When the change in the values of two variables is neither proportional nor fixed i.e. corresponding to a unit change in one variable, the other variables change at a different rate, then the variables are said to be *non-linear or curvilinearly correlated.*

If such data is plotted, then we get a curve shape but not a straight line.

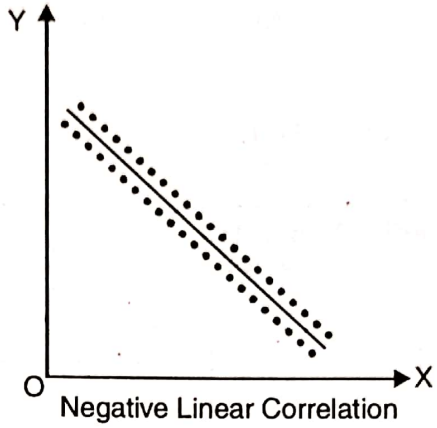
e.g.

X:	10	20	30	40	50	X:	8	9	9	10	10	28	29
Y:	40	60	80	100	120	Y:	80	130	150	170	200	250	360
Linear Correlation						Non-linear Correlation							

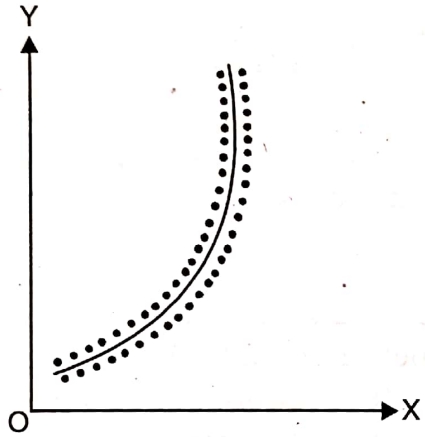
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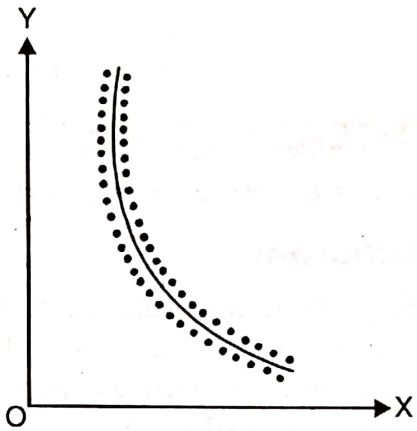
Positive Linear Correlation



Negative Linear Correlation



Positive Curvilinear Correlation

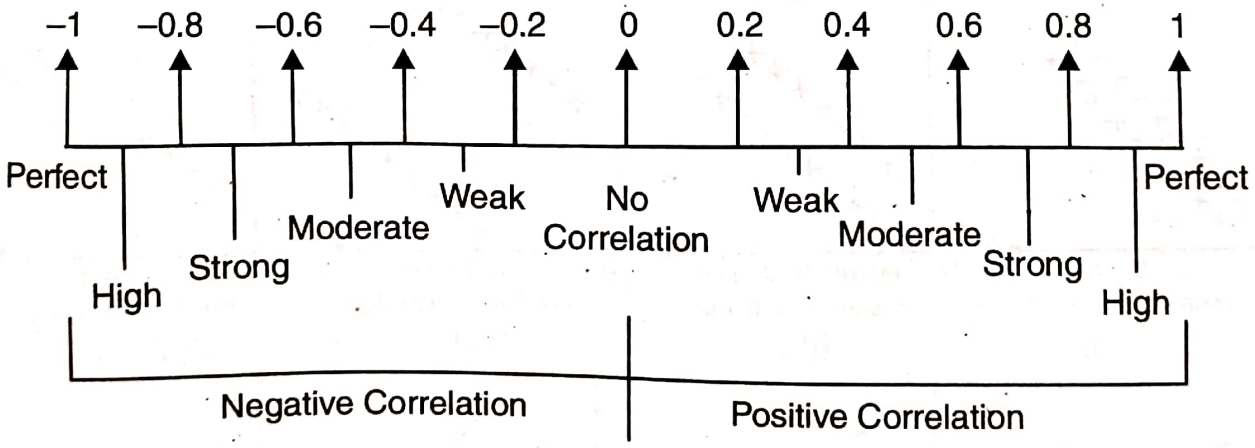


Negative Curvilinear Correlation

7.7 INTERPRETATION OF CORRELATION COEFFICIENT

The correlation coefficient (r) ranges from –1 (a perfect negative correlation) to 1 (a perfect positive correlation.)

$$i.e. -1 \leq r \leq 1$$



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1. KARL PEARSON'S COEFFICIENT OF CORRELATION

The *British statistician* **Prof. Karl Pearson** has devised the widely used mathematical method to measure numerically the degree and direction of the relationship between two linearly related variables. This method is popularly known as *Karl Pearson's Coefficient of Correlation or product Moment Method* or *Covariance method.* It is considered as the ideal measure of correlation.

Characteristics of Karl Pearson's Coefficient of Correlation :

1. **Ideal measure :** It is considered to be an ideal measure of correlation as it is based on covariance and variance of X and Y, which are most reliable standard statistical tools.
2. **Determines the direction of relationship :** It establishes the direction of relationship between variables *i.e.,* positive or negative.
3. **Determines the degree of relationship :** It is a quantitative method, so measure the size (degree) of relationship between the variables. The coefficient of correlation ranges between –1 and +1, where +1 means perfect +ve correlation, –1 means perfect –ve correlation and 0 means no correlation between the variables.
4. **Based on arithmetic mean and standard deviation :** Its formula is based upon arithmetic mean and standard deviation, which are most reliable statistical tools.

Definition of Karl Pearson's Coefficient of Correlation

Definition : A ratio between covariance of two variables to the product of their standard deviations is called **Karl Pearson's Coefficient of Correlation.** It is denoted by symbol r or $r(x, y)$.

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